# Cloudie

CSP650 · CYBERSECURITY

# Secure cloud storage that reads how a file is built

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THE IDEA

## Signatures know the past. Structure reads the file.

Antivirus signatures only catch files someone has already seen. Cloudie also checks **how a file is put together**, so a disguised file is caught even when no engine has a signature for it yet. Every file is then encrypted before it is stored.

THE PROBLEM

**01 New malware has no signature yet**
Signature scanners miss files nobody has reported.

**02 Attackers disguise files**
A macro inside a .docx, `invoice.pdf.exe`, a ZIP hidden in a picture.

**03 Stored files can leak**
Files kept unencrypted are exposed if storage is breached.

HOW IT WORKS

## Every upload is scanned, judged and encrypted before it is stored

**Upload**
Type allow-list, 25 MB limit, rate limit

**VirusTotal**
70+ engines, primary verdict
FAIL CLOSED

**Structure checks**
PDF, Office, ZIP, CSV, images, file name
NO SIGNATURE

**AI for PDFs**
Random Forest, 32 features, explained
EXPLAINABLE

**AES-256**
New key + IV per file, key goes to you

**Cloud storage**
Ciphertext only, on Google Cloud

INNOVATION · STRUCTURE LAYER

## What the file is built from, by file family

| FAMILY | WHAT THE RULES CATCH | MITRE ATT&CK |
|---|---|---|
| **PDF** | Hidden keywords (`/J#61vaScript`), exploit code in scripts, embedded programs | T1027 · T1059.007 |
| **Office** | Macro inside a macro-free .docx, DDE, remote template | T1059.005 · T1221 · T1559.002 |
| **ZIP** | Zip slip (`../`), zip bomb, programs disguised as documents | T1036.007 · CWE-22 |
| **CSV** | Cells that run commands (`=cmd|…`) or leak data | T1559.002 · CWE-1236 |
| **Images** | A ZIP, program or script hidden after the picture ends | T1027.003 |
| **Every file** | Real type vs. its name, double extension, right-to-left trick | T1036.008 · T1036.002 |

Bytes are only read, never run or extracted, with a 3-second budget per file.

VERDICT ENGINE

## VirusTotal decides first

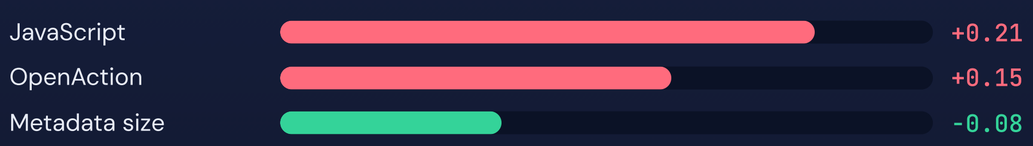
| | |
|---|---|
| 2 or more engines flag it | BLOCKED |
| No answer from VirusTotal | BLOCKED |
| Only 1 engine (classic false positive) | SUSPICIOUS |
| Clean, but a strong structure finding | SUSPICIOUS |
| Clean, but AI flags a risky PDF | SUSPICIOUS |
| Everything clean | CLEAN |

Suspicious files are stored encrypted and tagged, with the reason shown in the scan report.

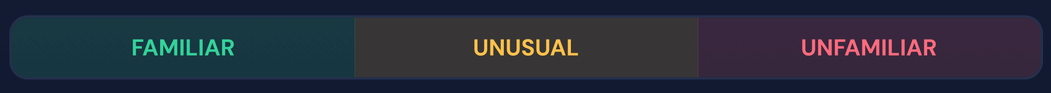
EXPLAINABLE AI

## The report says why, and when it is unsure

**Why this AI score:** each feature's exact contribution from the trees.




**Familiarity check:** how close is this PDF to the PDFs it was trained on?




Bars show an example layout, not a measured file.

PROTECTING WHAT IS STORED

## Only the owner holds the key

**Key not stored**
Only a SHA-256 hash of each file's key is kept

**Integrity check**
Decrypted bytes must match the hash taken at upload

**Sealed audit log**
HMAC hash chain: edits and deletions show up

**2-step login**
Password + one-time code, rate limited

**Private scanner**
Only the web service may call it (Cloud IAM)

**QR key**
Key delivered by email and QR, generated on our server

EVIDENCE · MEASURED, NOT ASSUMED

| | |
|---|---|
| **99.87%** | Hold-out accuracy on 18,126 PDFs, 0 false positives |
| **99.46%** | Harder test: trained on small PDFs, tested on the largest |
| **1.0%** | Real benign PDFs tagged (2 of 200), 0 blocked |
| **256** | Automated tests (81 PHP + 175 Flask) |

STANDARDS · MAPPED, NOT CERTIFIED

| | |
|---|---|
| MITRE ATT&CK | 12 techniques the checks detect |
| NIST CSF 2.0 | Identify · Protect · Detect · Respond · Recover |
| ISO/IEC 27001 | Annex A 8.5, 8.7, 8.13, 8.15, 8.24 |
| OWASP | File Upload Cheat Sheet, Top 10 |
| NIST 800-63B | Password and login rules |

LIVE NOW · TRY IT YOURSELF

## Scan, sign up, upload a file.

Runs on Google Cloud Run, Cloud SQL and Cloud Storage. For individuals, SMEs, schools, NGOs and government offices.

UN SDG 9 · Industry, Innovation & Infrastructure

UN SDG 16 · Strong Institutions

Live app

Source code